

5-3 Project One Submission: Virtual Systems and Networking Concept Brief

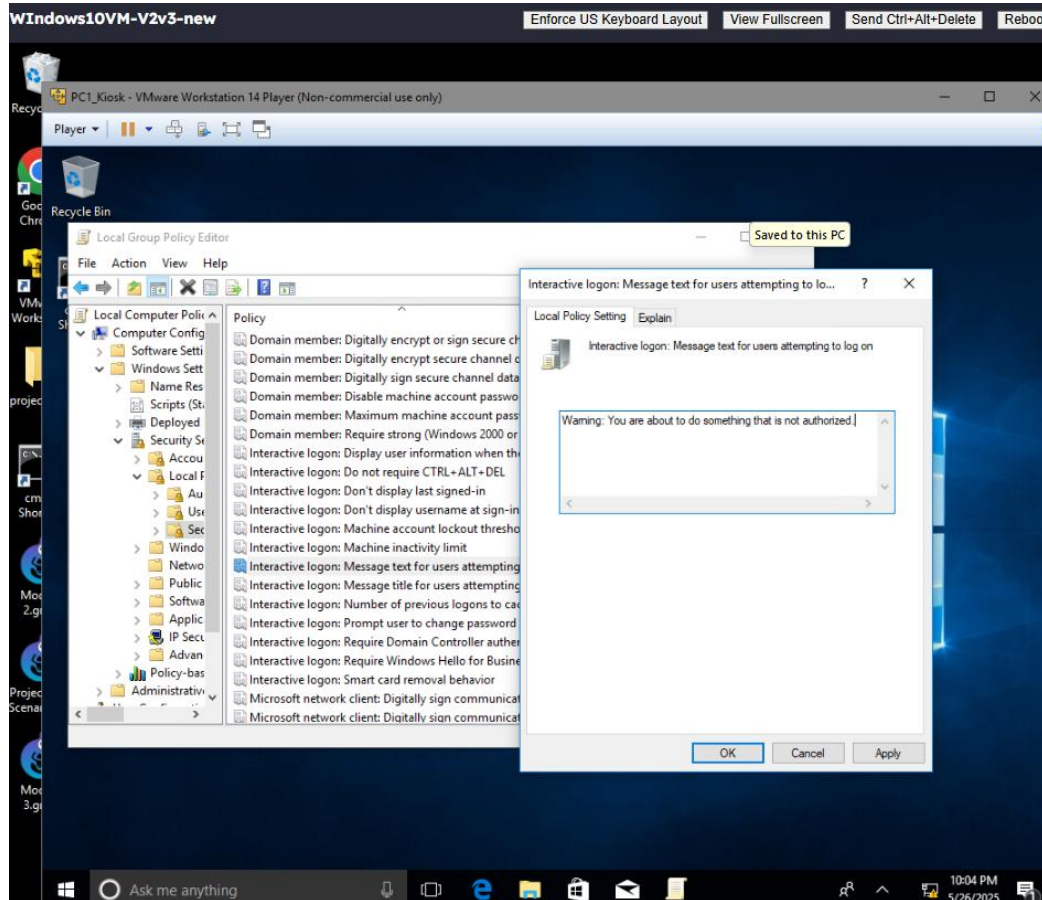
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CYB 220



1. Using Virtual Systems

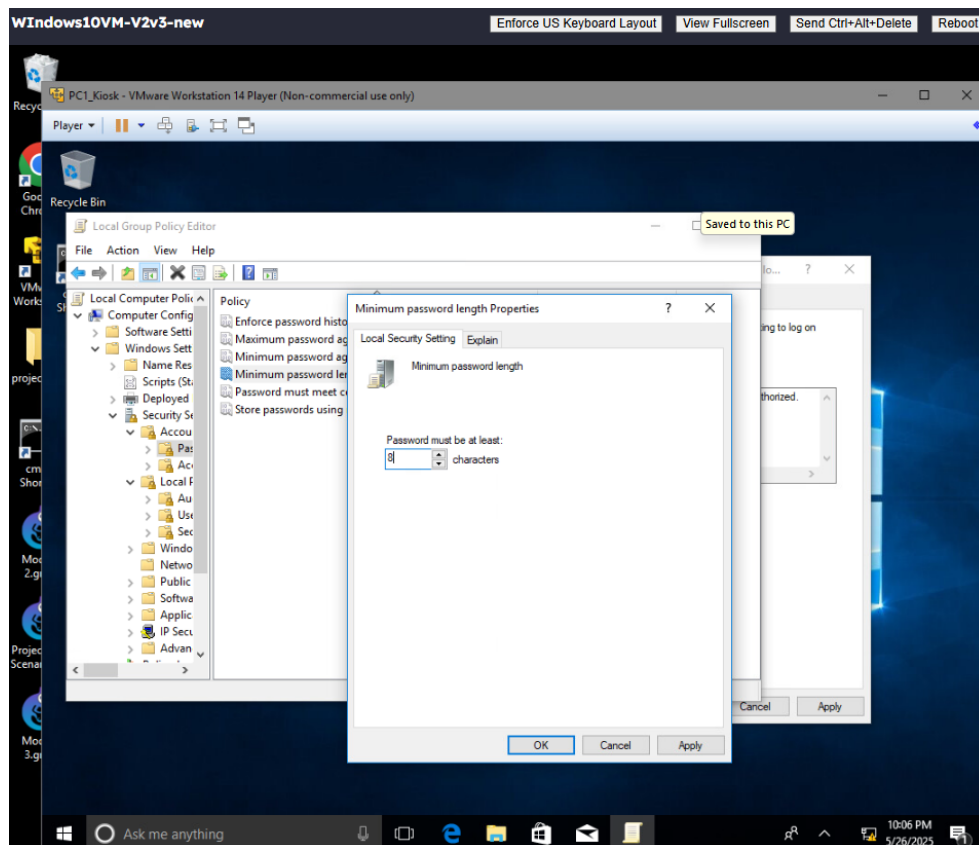
A. Configure group policy **settings** to meet project specifications as identified in the InfoSec Project One Activity. Provide screenshots from the local group policy editor dialog box as evidence of each change.

- Set the **Interactive logon: Message text for users attempting to log on** prompt to “Warning: You are about to do something that is not authorized.”



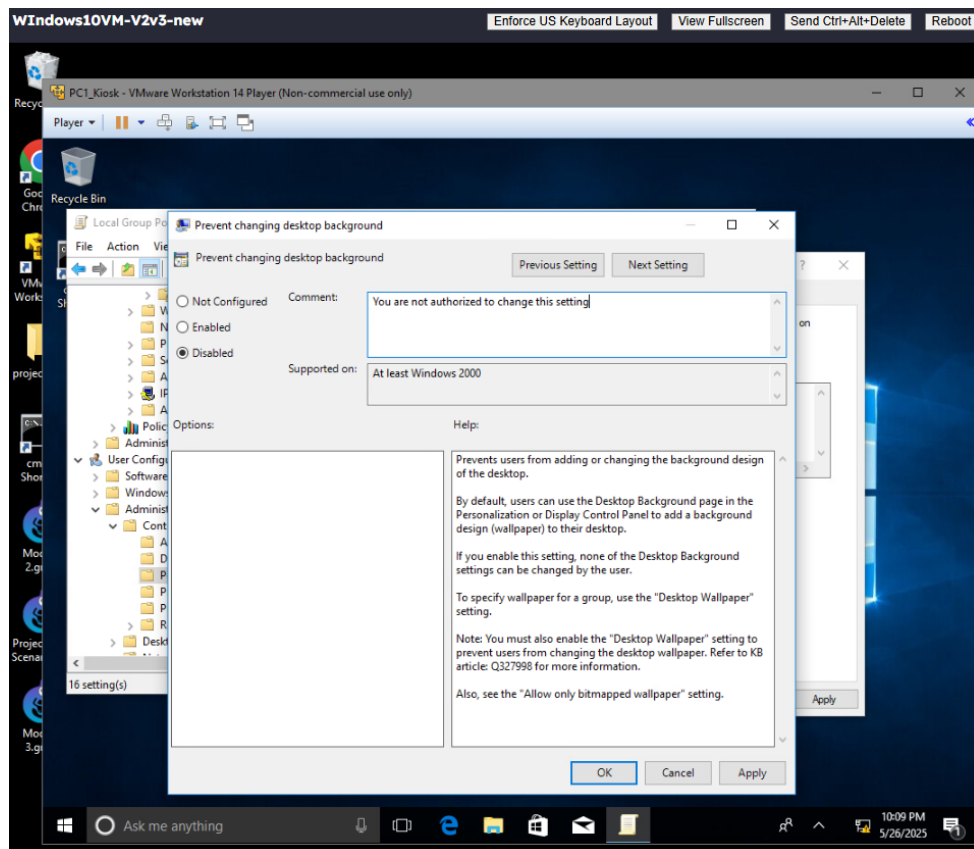
•Change local password policy setting.

•Change the minimum password length to 8.



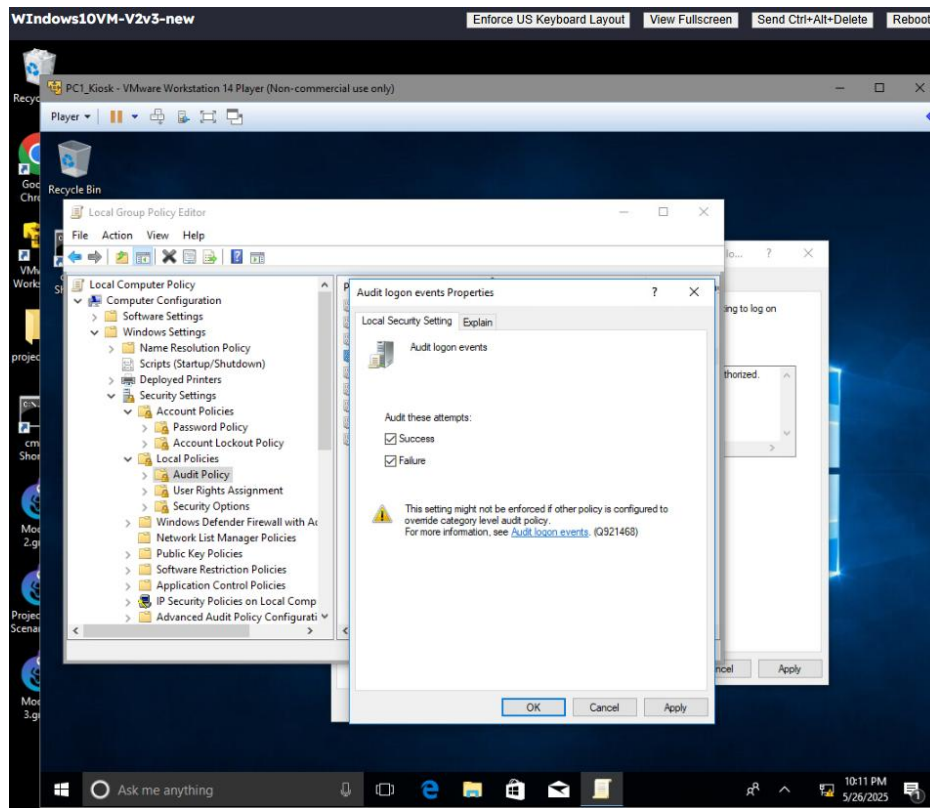
•Change desktop background user rights assignment (disable non-admin change capability).

•Set **Prevent changing desktop background** to disabled. Add the comment "You are not authorized to change this setting" to the comment box.



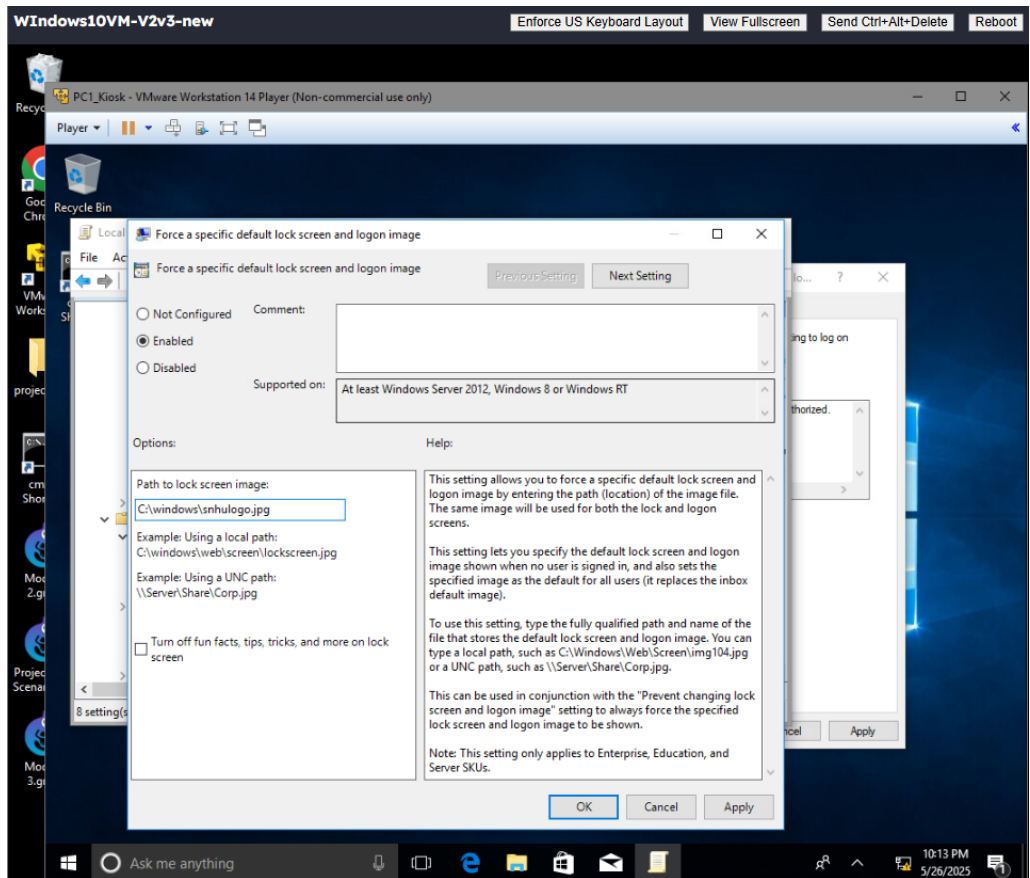
•Configure local audit policy setting.

•Set **Audit logon events** to enabled.



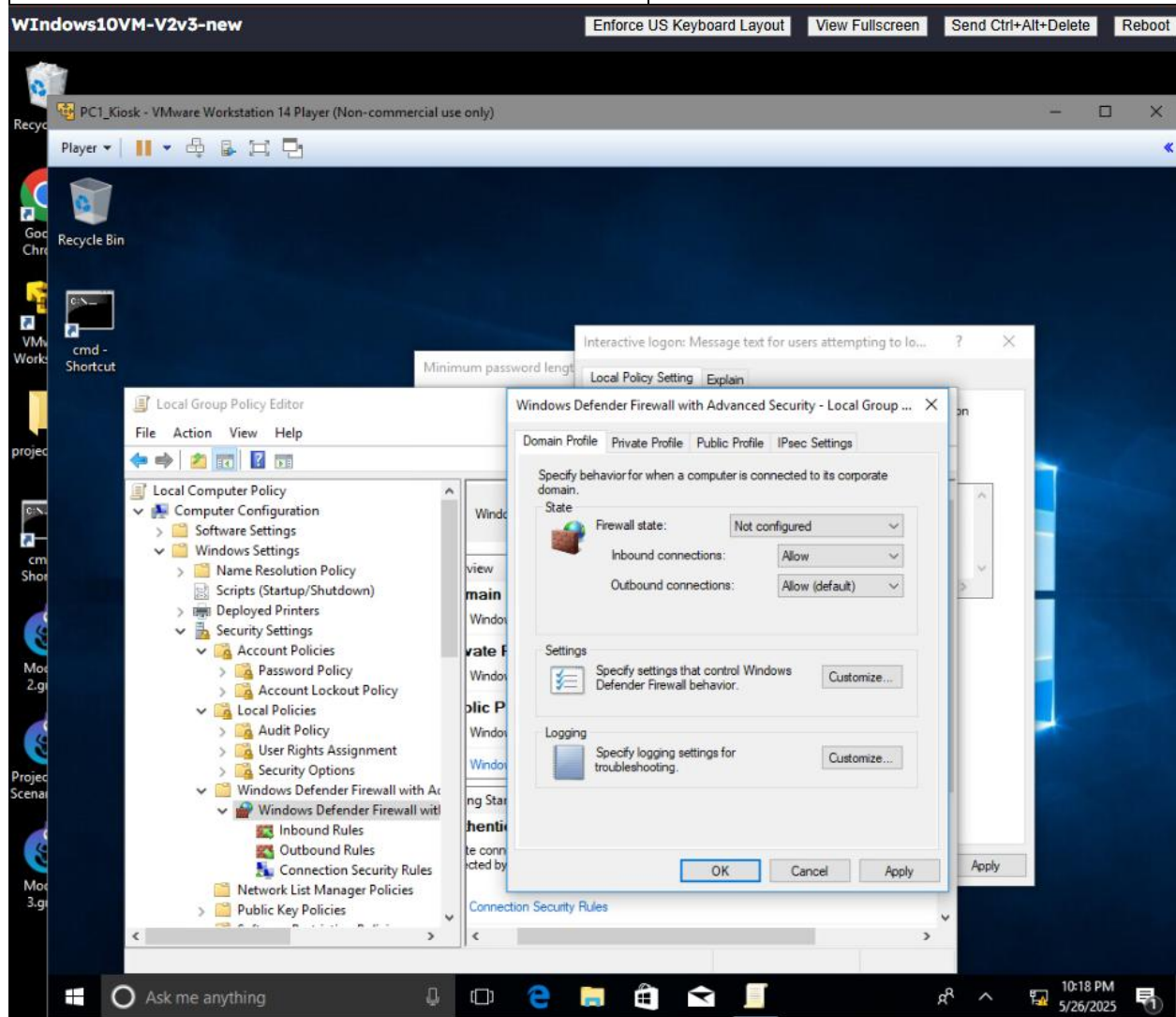
- Configure default logon banner (warning that requires direct affirmation to continue).

- Set Force a specific default lock screen and logon image to enabled, and provide a **path to the lock screen image** of C:\windows\SNHULogo.jpg.



- Change default Windows firewall profile.

- Set Windows Defender Firewall Properties\Firewall State to on, and set **Inbound connections** and **Outbound connections** to allow.

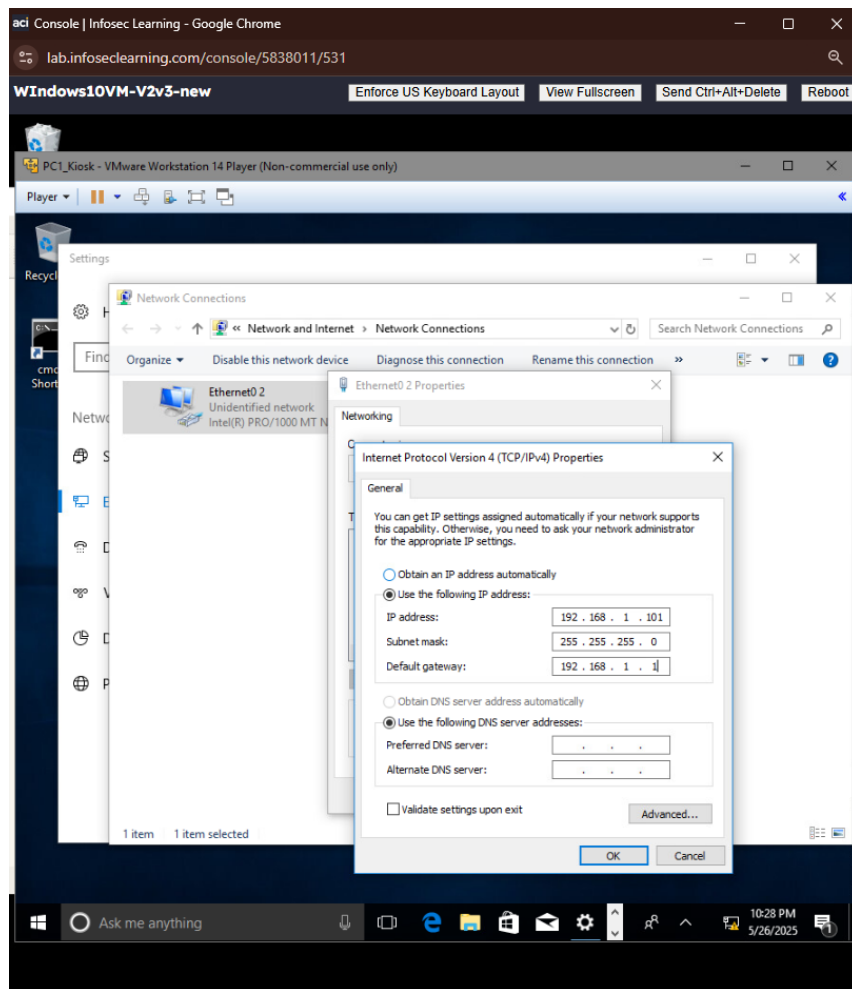


- B.** Configure **network** resources to meet the network configuration settings as identified in the InfoSec Project One Activity. Provide a screenshot of each network setting configuration.

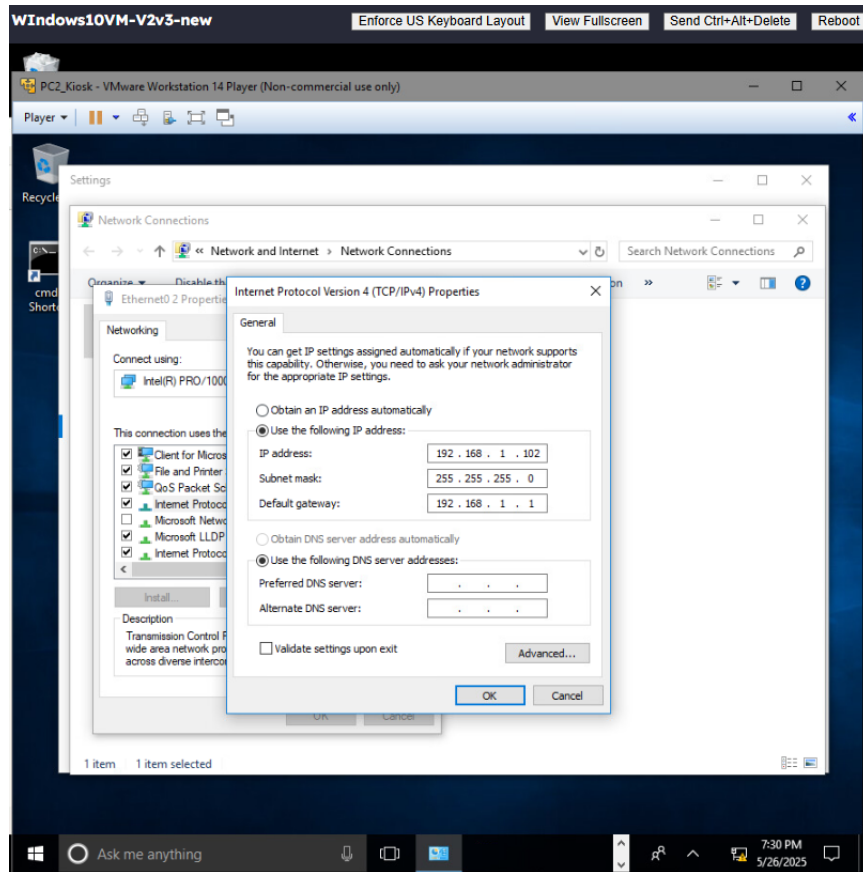
Network Router Setup

```
WIndows10VM-V2v3-new Enforce US Keyboard Layout View Fullscreen Send Ctrl+Alt+Delete Reboot
Network_Router
et2/0, changed state to down
Network_Router#ipconfig
Translating "ipconfig"
Translating "ipconfig"
% Unknown command or computer name, or unable to find computer address
Network_Router#ip a
^
% Invalid input detected at '^' marker.
Network_Router#ip /a
^
% Invalid input detected at '^' marker.
Network_Router#
Network_Router#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
Network_Router(config)#int fa0/0
Network_Router(config-if)#ip address 192.168.1.1 255.255.255.0
Network_Router(config-if)#no shutdown
Network_Router(config-if)#
*Mar 1 00:25:23.611: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:25:24.611: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, c
hanged state to up
Network_Router(config-if)#exit
Network_Router(config)#int fa0/1
Network_Router(config-if)#ip address 192.168.2.1 255.255.255.0
Network_Router(config-if)#no shutdown
Network_Router(config-if)#
*Mar 1 00:26:11.063: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:26:12.063: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, c
hanged state to up
Network_Router(config-if)#exit
Network_Router(config)#Andree Salvo
```

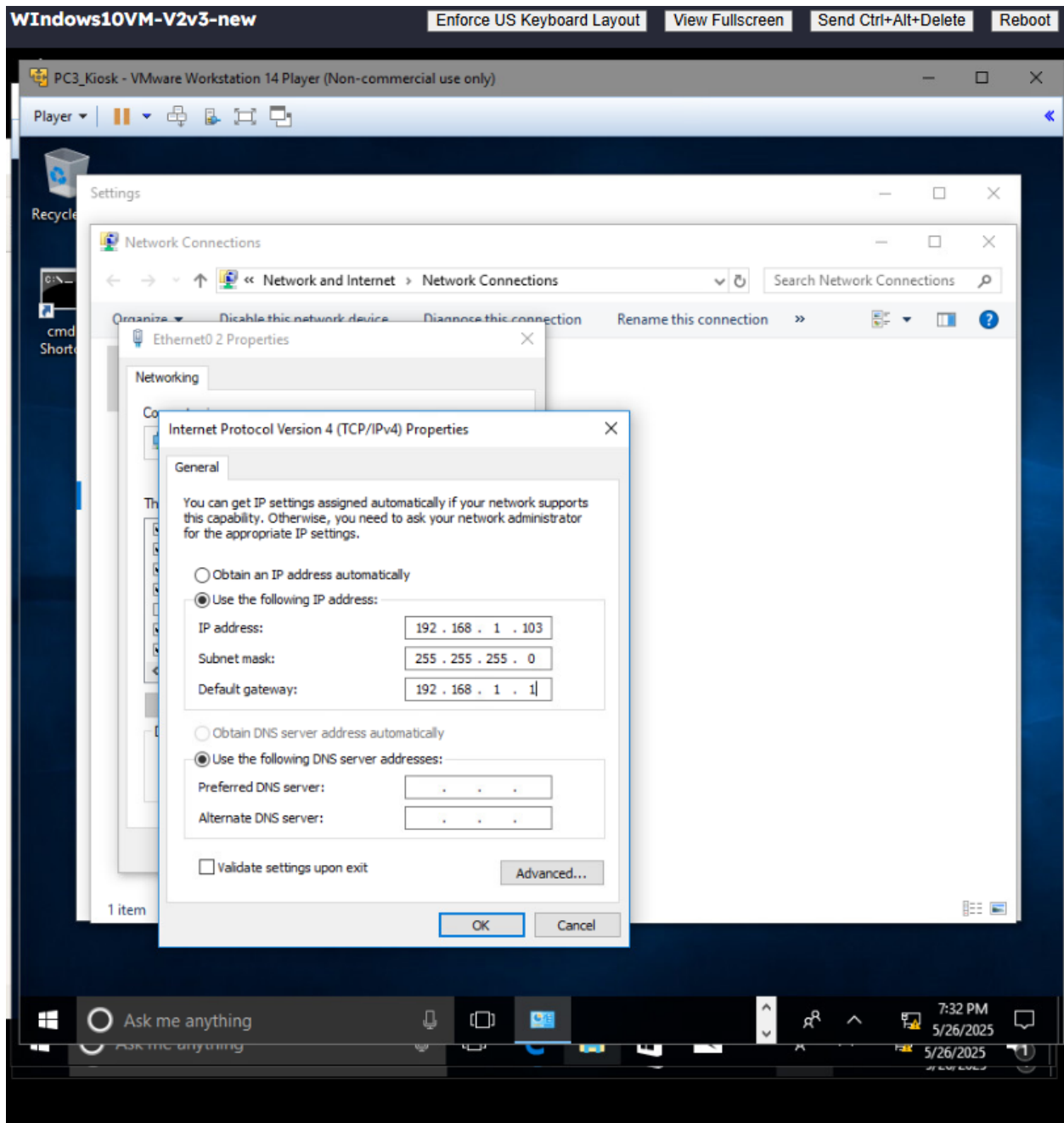
PC1_Kiosk Setup



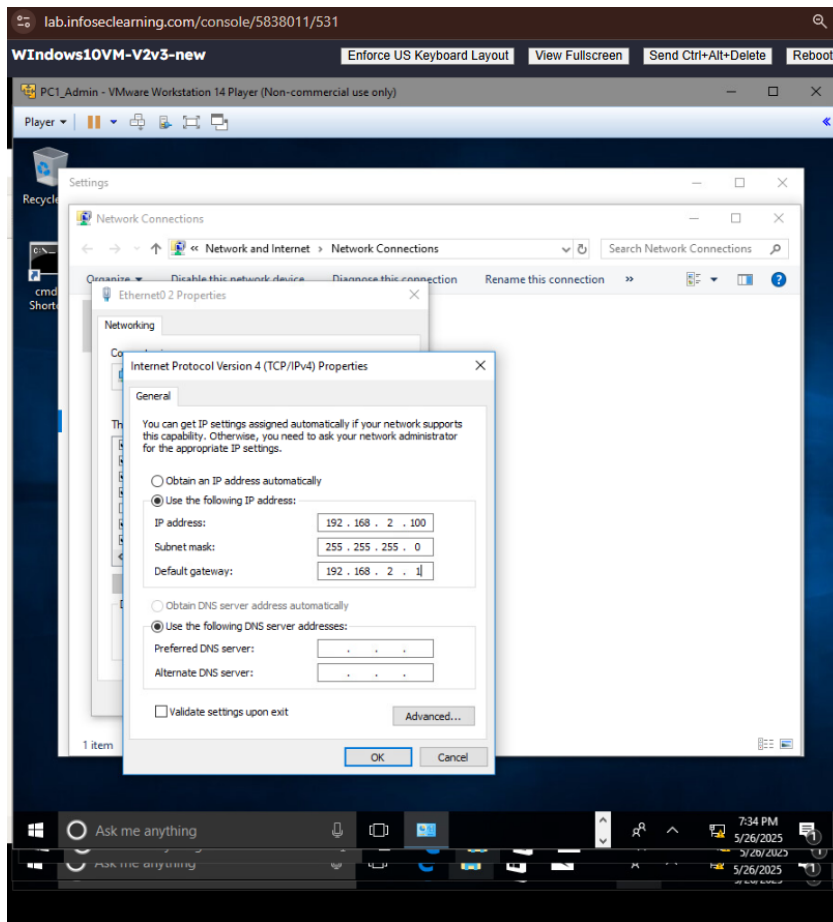
PC2_Kiosk Setup



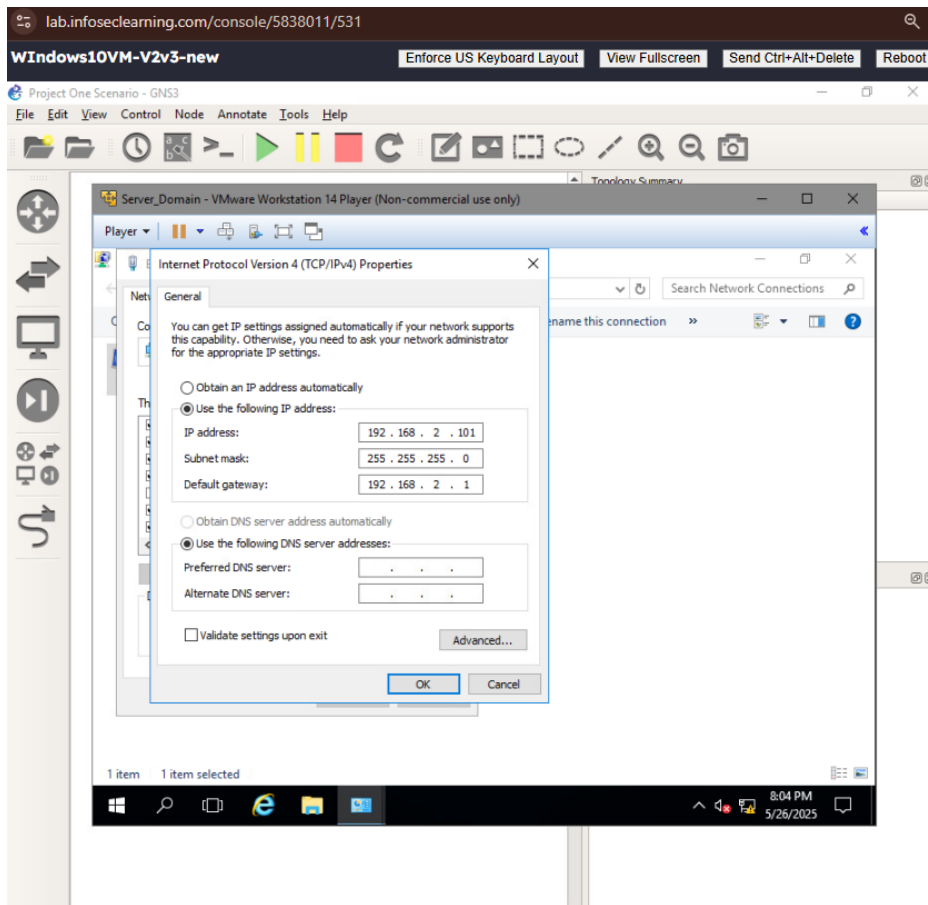
PC3_Kiosk Setup

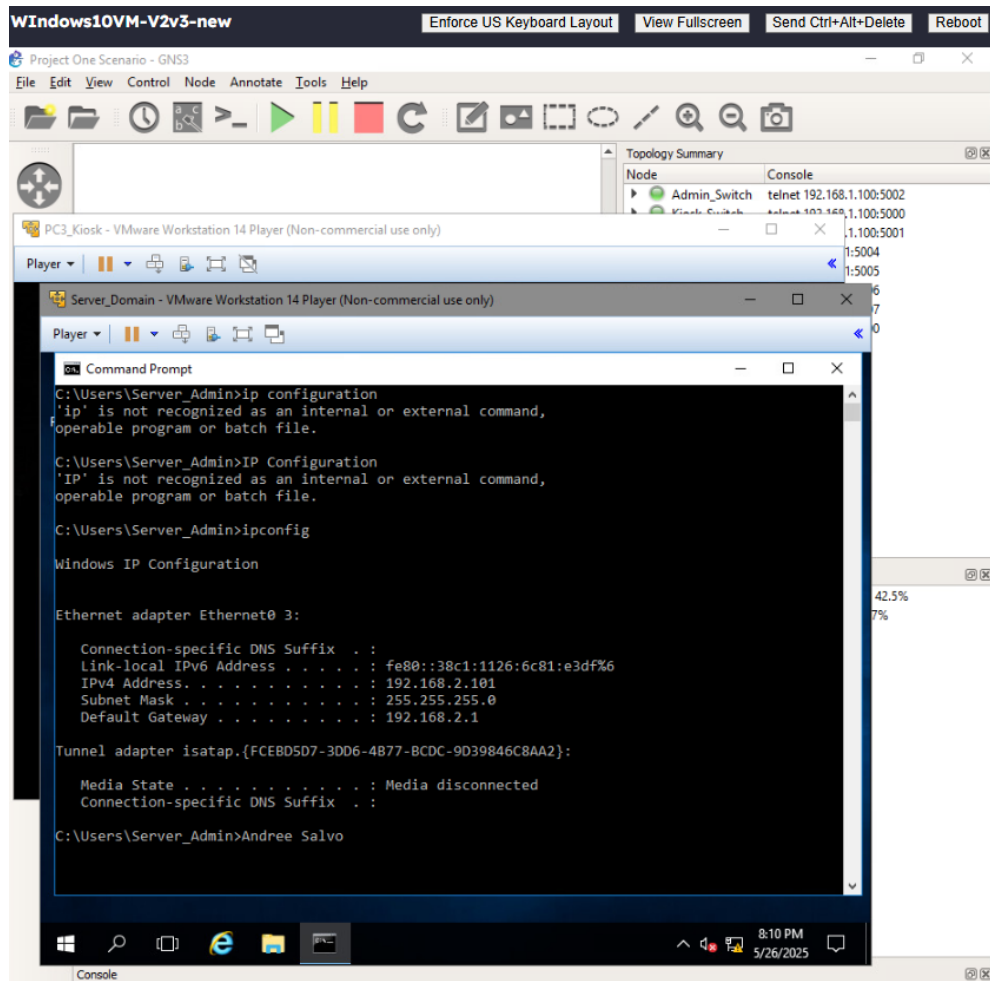


PC1_Admin Setup



Server_Domain Setup





I. **Conceptual Justification for Uses of Virtual Systems**

A. **Describe the benefits of using virtualization technology in sandboxing.**

Answer: Benefits of using a virtualization technology in sandboxing include things like enhanced security, which “isolates untrusted or potentially malicious code within a controlled environment that would prevent unauthorized users from gaining access to sensitive data” (Paubox, 2021), safe testing and development, improved stability where when you do use sandboxing it can help the systems stability by help preventing any failures.

B. **Describe the drawbacks of using virtualization technology in sandboxing.**

Answer: The Drawbacks that I personally encountered were Performance. When I was using virtualization, I remembered that I had multiple Virtual machines (VMs) running, and my computer started responding very slowly and lagging a little bit, but another drawback of using virtualization is vulnerabilities. If the virtualization isn't properly installed, malware can exploit and escape inside the VM, causing it to become vulnerable to the entire system. And if the exploit does successfully escape, it can get inside your main host PC.

C. **Describe a situation other than sandboxing where virtualization technology would be a valuable solution.**

Answer: A situation besides sandboxing, where virtualization would be a valuable solution, would have to be a Virtual disaster recovery. A virtual disaster recovery is most efficient where organizations replicate VMs to another location, which keeps a live backup ready if something does go wrong. The benefits of using VDR are that “they are containerized into VMs, independent from the underlying hardware. An organization does not need the same physical servers at the primary site as at its secondary disaster recovery site.” (Crocetti, 2022)

References:

Paubox. (2021, October 12). *Benefits of sandboxing*. Retrieved from
<https://www.paubox.com/blog/benefits-of-sandboxing>

Crocetti, P. (2022, December 16). *Virtual disaster recovery*. TechTarget.
<https://www.techtarget.com/searchdisasterrecovery/definition/virtual-disaster-recovery>